

Question 1

a) $\frac{1 + \sqrt{7}}{3 - \sqrt{7}} = a + b\sqrt{7}$

By rationalizing the denominator, find the values of a and b (2)

b) Factorise fully: $x^4 - ax^2 - 4x^2 + 4a$ (2)

c) $f(x) = \begin{cases} |x + 1| & \text{for } x \leq 0 \\ 4 - x^2 & \text{for } x > 0 \end{cases}$

Sketch the graph of $y = f(x)$ (3)

d) Find $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ if $f(x) = x^2 + 1$ (2)

e) Find the largest possible domain of

$$f(x) = \sqrt{x+2} - \sqrt{2-x}$$

(2)

-4

(2)

(1)

a) Expand $(x + y)^3$

(3)

b) Solve $\frac{x}{x+1} \geq 5$

c) For the curve $y = \frac{x^2}{1-x^2}$

(1)

(i) Write down the x and y intercepts

(1)

(ii) Find whether the function is odd, even or neither

(2)

(iii) Write down the equations of all asymptotes

(2)

(iv) Hence sketch the curve $y = \frac{x^2}{1-x^2}$

(1)

(v) What is the range of the function?

Question 3 (Start a new page)

a) Sketch $y = \cos x$ for $0^\circ \leq x \leq 360^\circ$ (1)

b) Find the exact value of $\cos 210^\circ$ (2)

c) Simplify: $\sin (180^\circ + \theta) \tan (90^\circ - \theta)$ (2)

d) Simplify: $(\cot^2 \theta + 1)(1 - \cos^2 \theta)$ (2)

e) Solve: $\cos x = -\frac{1}{\sqrt{2}}$ for $0^\circ \leq x \leq 360^\circ$ (2)

f) If $\sec \theta = \frac{7}{3}$ and $\sin \theta < 0$ find the exact value of $\tan \theta$ (2)

(4)

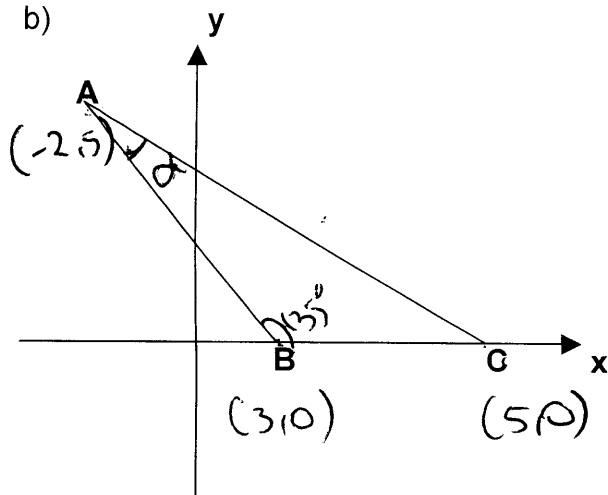
Question 4 (Start a new page)

Marks

- a) Find the coordinates of the point which divides the interval LM in the ratio 2:3 given L is $(-4, 1)$ and M is $(5, 6)$

(2)

b)



A is the point $(-2, 5)$
 B is the point $(3, 0)$
 C is the point $(5, 0)$

Diagram not to scale

- (i) Find the exact distance AC
- (ii) Find the gradient of AB
- (iii) Show that $\angle ABC = 135^\circ$
- (iv) Hence find $\angle BAC$ to the nearest minute

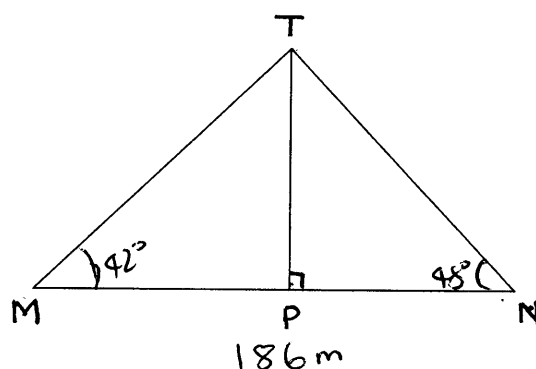
(1)

(1)

(1)

(2)

c)



The top, T , of a radio mast TP has an angle of elevation of 42° from a point M on the level ground on which it stands, and of 48° from a point N on the other side directly opposite M

If MN is 186 metres find the height of the radio mast TP .

(4)

QUESTION 1

$$a) \frac{1+\sqrt{7}}{3-\sqrt{7}} \times \frac{3+\sqrt{7}}{3+\sqrt{7}} = \frac{3+\sqrt{7}+3\sqrt{7}+7}{9-7}$$

$$= \frac{10+4\sqrt{7}}{2}$$

$$= 5+2\sqrt{7}$$

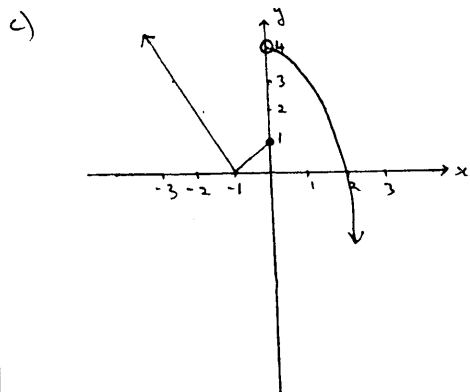
$$\therefore a=5, b=2$$

$$b) x^4 - ax^2 - 4x^2 + 4a$$

$$= x^2(x^2 - a) - 4(x^2 - a)$$

$$= (x^2 - a)(x^2 - 4)$$

$$= (x^2 - a)(x-2)(x+2)$$



$$d) \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{[(x+h)^2 + 1] - [x^2 + 1]}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + 1 - x^2 - 1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(2x+h)}{h}$$

$$= \lim_{h \rightarrow 0} (2x+h)$$

$$= 2x$$

Marks

$$e) x+2 \geq 0 \quad 2-x \geq 0$$

$$x \geq -2 \quad -x \geq -2$$

$$x \leq 2$$

$$\text{Domain is: } -2 \leq x \leq 2$$

QUESTION 2

$$a) (x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$$

$$b) \frac{x}{x+1} \geq 5$$

$$(x+1)^2 \frac{x}{x+1} \geq 5(x+1)^2 \quad x \neq -1$$

$$x(x+1) \geq 5(x+1)^2$$

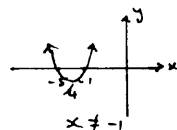
$$0 \geq 5(x+1)^2 - x(x+1)$$

$$0 \geq (x+1)(5(x+1) - x)$$

$$0 \geq (x+1)(5x+5-x)$$

$$0 \geq (x+1)(4x+5)$$

$$-\frac{5}{4} \leq x < -1$$



$$c) y = \frac{x^2}{1-x^2}$$

$$i) \text{ if } x=0, y=0 \therefore (0,0)$$

$$ii) f(-x) = \frac{(-x)^2}{1-(-x)^2}$$

$$= \frac{x^2}{1-x^2}$$

$$\therefore \text{even function}$$

$$ii) 1-x^2 \neq 0$$

$$(1-x)(1+x) \neq 0$$

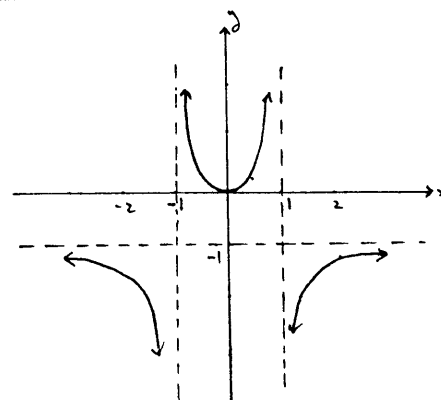
$$\text{vertical asymptotes: } x=1$$

$$x=-1$$

$$\lim_{x \rightarrow \infty} \frac{\frac{x^2}{1-x^2}}{\frac{x^2}{x^2}} = \frac{1}{0-1}$$

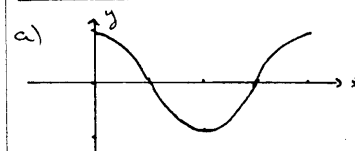
$$= -1$$

$$\text{Horizontal asymptote: } y=-1$$



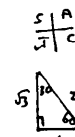
$$v) \text{Range: } y \geq 0 \quad y < -1$$

QUESTION 3



$$b) \cos 210^\circ = -\cos 30^\circ$$

$$= -\frac{\sqrt{3}}{2}$$



$$c) \sin(180^\circ + \theta) + \tan(90^\circ - \theta)$$

$$= -\sin \theta + \cot \theta$$

$$= -\sin \theta + \frac{\cos \theta}{\sin \theta}$$

$$= -\cos \theta$$

$$d) (\cot^2 \theta + 1)(1 - \cos^2 \theta)$$

$$= \csc^2 \theta \cdot \sin^2 \theta$$

$$= \frac{1}{\sin^2 \theta} \cdot \sin^2 \theta$$

$$= 1$$

Marks

$$e) \cos x = -\frac{1}{\sqrt{2}}$$

$$x = 180^\circ - 45^\circ, 180^\circ + 45^\circ$$

$$x = 135^\circ, 225^\circ$$

$$f) \frac{s}{T} = \frac{A}{C}$$

$$\frac{5}{7} = \frac{A}{C}$$

$$4^{\text{th}} \text{ quadrant}$$

$$x^2 = 7^2 - 3^2$$

$$= 40$$

$$x = \sqrt{40}$$

$$= 2\sqrt{10}$$

$$\therefore \tan \theta = -\frac{2\sqrt{10}}{3}$$

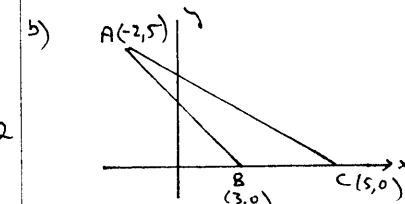
QUESTION 4

$$a) \frac{kx_1 + ly_1}{k+l}, \frac{kx_2 + ly_2}{k+l}$$

$$= \left(\frac{2 \times 5 + 3 \times 4}{2+3}, \frac{2 \times 6 + 3 \times 1}{2+3} \right)$$

$$= \left(\frac{10+12}{5}, \frac{12+3}{5} \right)$$

$$= \left(\frac{22}{5}, 3 \right)$$



$$b) AC = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(5 - (-2))^2 + (0 - 5)^2}$$

$$= \sqrt{7^2 + 5^2}$$

$$= \sqrt{74}$$

$$ii) m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{0 - 5}{3 - -2}$$

$$= \frac{-5}{5}$$

$$= -1$$

$$iii) m = \tan \theta$$

$$-1 = \tan \theta$$

$$\theta = 180 - 45$$

$$= 135^\circ$$

$$\therefore \angle ABC = 135^\circ$$

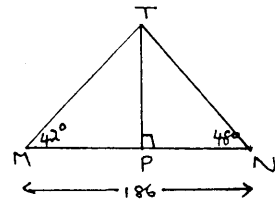
$$iv) BC = 2 \text{ units}$$

$$\frac{\sin \angle BAC}{2} = \frac{\sin 135}{\sqrt{74}}$$

$$\sin \angle BAC = \frac{\sin 135}{\sqrt{74}} \times 2$$

$$\angle BAC = 9^\circ 28'$$

c)



$$\angle MTP = 48^\circ$$

$$\angle NTP = 42^\circ$$

$$\therefore \angle MTN = 90^\circ$$

$$\cos 48^\circ = \frac{TN}{186}$$

$$TN = 124.4582928$$

$\therefore$ In $\triangle TPN$

$$\sin 48^\circ = \frac{TP}{124.4582928}$$

$$TP = 92.49 \text{ metres}$$

Marks

1

1

2

4

Marks